

Welcome to ML4CI

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July 2025

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1 Welcome to Machine Learning for Causal Inference

Welcome to ML4CI!

In this short course, we will spend the next few days learning about the basics behind using machine learning methods for estimating cause-effect relations.

By the end of this course you should have a solid understanding of:

- the challenges of estimating causal effects with data
- when/why one should use machine learning to estimate causal effects
- the challenges of using machine learning methods for estimating causal effects
- how to estimate causal effects with machine learning methods in R

The literature on ML and causal inference is very expansive and complex. There are some key essential concepts that are not easy to understand for those with little to no formal technical background. The purpose of this course is to help you with your efforts in filling this gap.

2 Overview of Course Topics

In this course, we will cover the following topics:

- Introduction to the Datasets
- Potential Outcomes, Estimands, Identifiability
 - ATE, ATT, and CATEs
 - Identification Bias versus Estimation Bias
- Parametric Regression for Effect Estimation
 - G Computation
 - Inverse Probability Weighting
- Double Robust Methods
 - Augmented Inverse Probability Weighting (AIPW)
 - Targeted Minimum Loss-Based Estimation (TMLE)
 - Introduction to Sample Splitting, Cross Fitting, and Cross Validation

- Conditional Average Treatment Effect Estimation
 - T, S, X, and R Learners
 - Causal Forests
 - The DR Learner
- Meta Learners for the Exposure and Outcome Models: Stacking
 - Fitting the Outcome and Propensity Score Models
 - The Super Learner: Intuition and a Worked Example
 - Screening Algorithms*
 - Design Matrices for the Super Learner*
 - Missing Data*
 - The Cross-Validated Super Learner
- Estimating Effects in Example Datasets
 - TMLE + SuperLearner
 - AIPW + SuperLearner
 - AIPW + Cross-Validated SuperLearner
 - CATE Estimation with the DR Learner

3 The Structure of the Course

Over the next week, we'll have the opportunity to go over the important concepts behind using machine learning for estimating causal effects.

We will also engage in a few "Workshop Exercises". In these exercises, we will take some time to give you the opportunity to write code to complete specific tasks. These may include:

- Fitting the Super Learner to obtain a dose-response function
- Fitting the DR Learner to estimate conditional average treatment effects
- Fitting an AIPW and TMLE estimator to estimate average treatment effects in real data

4 R Code Logistics

In this short course, we will be using the R programming language throughout. In order to run the code we will be using during the course of the workshop, it will be important to have a few packages pre-installed on your computer.

The easiest way to install these packages would be to paste the following code in the R console¹ and run it:

```
pacman::p_load(tidyverse, here, sandwich, lmtest,
               boot, ranger, ggplot2, broom, SuperLearner,
               tmle, AIPW, ranger, xgboost, e1071,
               nnet, glmnet, remotes)

remotes::install_github("tlverse/tlverse")
library(tlverse)
```

¹ For the beginner, I highly recommend using RStudio as your IDE of choice, particularly if you are not familiar with R and R programming. There are many excellent resources for installing and setting up R and the RStudio IDE. Here is a good getting started guide, provided by Garrett Grolemond: <https://rstudio-education.github.io/hopr/starting.html>

If the above code does not work for you, consider installing each package separately:

```
install.packages("tidyverse")
library(tidyverse)
install.packages("here")
library(here)
install.packages("sandwich")
library(sandwich)
...
install.packages("remotes")
library(remotes)

remotes::install_github("tlverse/tlverse")
library(tmle3)
library(sl3)
```

4.1 Using devtools instead of remotes

Sometimes, installing development packages (i.e., using `install_github()`) can lead to installation errors and problems. If you encounter these errors, the best recommendation I can give is to try using `devtools` instead of `remotes`:

```
install.packages("devtools")
library(devtools)

devtools::install_github("tlverse/tlverse")
library(tlverse)
library(s13)
```

Note that, as of 2024-08-02, the `remotes` package is still available and being maintained. However, there may soon be a replacement package known as `pak`.²

² <https://pak.r-lib.org/>

4.2 Manually Installing Dependencies

If this doesn't work, the next option is to look carefully through the log at the error messages. Often, these errors arise because a certain dependency could not be installed. It would help to try to install those dependencies first, and then try again. This could mean sometimes installing compilation libraries on your computer first. For example,

installation of package 'igraph' had non-zero exit status

After running a successful `install.packages` for the `igraph` package, you could try installing `tlverse` again.

Sometimes, a dependency may no longer be available on the CRAN repository, which can create challenges for installing things in a straightforward way. For example, the `s13` package depends on the installation of the `imputeMissings` package, which (as of 2023-08-18) has been removed from CRAN.³ At this point, the maintainers of the `s13` package have to decide whether to remove `imputeMissings` as a dependency for their package. In the meantime, one solution to this problem is to install an archived version of the `imputeMissings` package. For example, one can run the following code⁴:

³ See: <https://bit.ly/46yYQ3R>

⁴ See: <https://bit.ly/3GhJTIT>

```
install.packages("https://cran.r-project.org/src/contrib/
  Archive/imputeMissings/imputeMissings_0.0.3.tar.gz")
```

Which should install the dependency from the CRAN archive. Once `imputeMissings` is successfully installed, one should be able to proceed with the `s13` installation as usual.

4.3 Addressing GitHub API Limits

There are two strategies one can pursue to deal with this error. First, the error arises because a single call to `install_github()` is attempting to install numerous packages at once. Without an authenticated API, this could easily reach the limit of request calls.

The easiest way to address this issue is to use a Github personal access token (PAT).

There are a number of ways to do this. Within R and RStudio, one straightforward way to manage PATs is to install and use the `usethis` package, which has a suite of functions available for creating and integrating PATs.

In the past, instructions were provided on how to add a PAT to your `.Renvirom` file. However, this practice is no longer recommended.⁵

⁵ See, for example: <https://bit.ly/41yJKKK>

- If you don't already have one, the first step is to create a GitHub PAT (see "Get a personal access token" in the practical instructions of the link provided in the margins).
- Once you have a token, copy it to your clipboard. You can then call the `gitcreds::gitcreds_set()` function, which should return a list of options to your RStudio console:

```
-> Your current credentials for 'https://github.com':

protocol: https
host    : github.com
username: ainaimi
password: <-- hidden -->

-> What would you like to do?

1: Abort update with error, and keep the existing credentials
2: Replace these credentials
3: See the password / token
```

Select "Replace these credentials" and paste you PAT in the allotted location.

Note that **your Github PAT is a password, and should be treated as such.**

Additionally, one can specify a time at which a particular PAT expires. If you're PAT is expired, simply follow the instructions [here](#) to renew them.

4.4 Addressing Other Issues

There are many other reasons why errors in installing development packages such as the `tlverse` can arise. Too many to cover here. The final route you can take, if it is available to you, is to contact the IT person at your school or workplace. These folks usually know how to resolve cryptic errors, especially those due to missing compilers.

4.5 An Additional Note on Software

If all else fails, and you are unable to install the `tlverse`, which includes the `tmle3` and `sl3` packages, most of what we'll be doing in the course can be accomplished with two alternative packages that are available on CRAN: the `tmle` package by Susan Gruber (Gruber and van der Laan, 2012), and the `SuperLearner` package by Eric Polley (Polley et al., 2016).

These packages are easier to install (directly from CRAN in the traditional way), and easier to use. But they do not have as many options for implementing in different settings, and do not support the `pipeline` architecture the way the `tlverse` does <https://tlverse.org/tlverse-handbook/>.

References

Susan Gruber and Mark J van der Laan. `tmle`: An r package for targeted maximum likelihood estimation. *Journal of Statistical Software*, 51(13):1–35, 2012.

Eric Polley, Erin LeDell, Chris Kennedy, and Mark van der Laan. *SuperLearner: Super Learner Prediction*, 2016. URL <https://github.com/ecpolley/SuperLearner>. R package version 2.0-22.